Charlie Johnson, PhD

Computational Biologist · Causal Inference & Real-World Oncology Data

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PROFILE

Computational biologist with 5+ years of industry experience at Tempus AI and a PhD from Imperial College London, completed while working full-time. My work sits where rigorous methods meet messy clinical reality: making causal and statistical inference reliable on real-world multimodal data (genomics, transcriptomics, EHR, claims, and imaging) and turning the results into decisions about biomarkers, indications, and trial design. I have led analyses for 15+ pharma and biotech partners across 25+ cancer types, and my doctoral research developed a causal framework for imaging biomarker development in lung cancer.

RESEARCH INTERESTS

Causal inference on real-world data · Biomarker discovery and validation · Multimodal integration of imaging, omics, and clinical data · Bias correction in observational studies · Precision oncology

PROFESSIONAL EXPERIENCE

Tempus AI — New York, NY / London, UK?

Senior Translational Scientist I — Sep 2025 – Present

Translational Scientist II — Feb 2023 – Sep 2025

Junior Translational Scientist — Jul 2021 – Feb 2023

Promoted twice in four years.

- **Partner science:** Lead data-driven research for 15+ pharmaceutical and biotech companies across 25+ cancer indications, building long-term partnerships that inform therapeutic development and biomarker-driven decisions.
- **Clinical strategy:** Analyse a multimodal oncology dataset of 300,000+ patients, integrating genomic, transcriptomic, and clinical data to guide biomarker selection and indication expansion for Phase 1 and 2 trials.
- **Methods:** Apply survival analysis, regression, differential gene expression, and unsupervised machine learning to shape partners' clinical strategy.
- **Real-world data:** Develop algorithms that address missing data, selection bias, and measurement bias, synthesising EHR, insurance claims, DNA-seq, and RNA-seq into analysis-ready evidence that was previously inaccessible.
- **Engineering:** Established coding and analytical best practices and built them into scalable pipelines and internal R packages used by other scientists, reducing time-to-insight [quantify].
- **Communication:** Translate high-level partner goals into scoped research projects, adapt to changing stakeholder priorities, and present findings to scientific and business audiences.
- **Leadership:** Train new hires and external clients on Tempus data pipelines; organise educational workshops as a member of the Tempus Board of Education.
- **Selected outputs:** analytics lead on Aadi Bioscience's real-world *TSC1/TSC2* characterisation study (AACR-NCI-EORTC 2023); *GNAS* colorectal cancer study (ASCO 2025).

EDUCATION

Imperial College London, Department of Surgery & Cancer — London, UK

PhD, Clinical Medicine Research — 20XX – 2026

- Completed part-time? alongside full-time employment at Tempus AI.
- Thesis: *A Causal Framework for Imaging Biomarker Development for Personalised Lung Cancer Management*.
- Supervisors: [names]
- [2–3 bullets: methods (causal approach, imaging modality, models), data (cohort / trial / registry, N), key findings]

New York University, Grossman School of Medicine — New York, NY

MS, Biomedical Informatics (GPA 3.89/4.00) — Jul 2021

- Thesis research: proteogenomic characterisation of endometrial carcinoma subtypes (Fenyö Lab).
- Coursework: Advanced Integrative Omics, Deep Learning, Machine Learning, Bioinformatics, Health Informatics, Methods in Quantitative Biology, Programming for Data Analysis.

Georgetown University, Georgetown College — Washington, DC

BS, Mathematics; Minor in Biology — May 2020

- Coursework: Ordinary Differential Equations, Linear Algebra, Abstract Algebra, Analysis I, Graph Theory, Biostatistical Methods, Data Visualization, Genetics, Biological Chemistry, Epidemiology & Public Health.

PUBLICATIONS

1. **Johnson I**, [co-authors] . [Title] . *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026.
2. Dou Y, Katsnelson L, Gritsenko MA, …, **Johnson I**, …, Fenyö D; Clinical Proteomic Tumor Analysis Consortium. Proteogenomic insights suggest druggable pathways in endometrial carcinoma. *Cancer Cell*. 2023;41(9):1586–1605.e15. doi:10.1016/j.ccell.2023.07.007

CONFERENCE PRESENTATIONS & ABSTRACTS

1. [Authors. Title] . *European Causal Inference Meeting (EuroCIM)*, 2026. Poster.
2. [Authors. Title] . *AI in Healthcare* [full name] , 2026. Poster.
3. [Authors] . Clinicogenomic landscape of metastatic colorectal cancer patients with pathogenic *GNAS* variants and associated real-world response rate and overall survival outcomes to oxaliplatin-based first-line treatment. *ASCO Annual Meeting*, 2025. Poster.
4. [Authors] . Real-world characterization and frequency of *TSC1* and/or *TSC2* alterations collected from tumor tissue and liquid biopsies from the Tempus genomic database in patients with advanced cancer. *AACR-NCI-EORTC International Conference on Molecular Targets and Cancer Therapeutics*, 2023. *Mol Cancer Ther* 22(12 Suppl):B003. Tempus analytics lead.
5. **Johnson I**, Villano A, Marshall MB. Design of a synthetic blood substitute for thoracic surgery simulation. *General Thoracic Surgical Club Annual Meeting*, Tucson, AZ, March 2018. Poster.

EARLIER RESEARCH EXPERIENCE

Fenyö Lab, NYU Grossman School of Medicine — New York, NY
Research Analyst — Dec 2020 – Jul 2021

- Characterised molecular subtypes of endometrial carcinoma from CPTAC/TCGA proteomic and genomic data in R and Python, contributing to the CPTAC *Cancer Cell* 2023 study.
- Showed that *CTNNB1* and *TP53* mutations drive significant changes in protein and phosphoprotein levels, with *CTNNB1* mutations linked to worse outcomes in low-grade tumours.
- Applied ICA (with t-SNE and silhouette-width cluster validation), WGCNA for gene modules and driver genes, and GSEA, and built a unified pipeline for target-gene investigation.

Maillard Lab, Georgetown University — Washington, DC
Lab Assistant — Jun 2017 – Jun 2018

- Prepared DNA samples and ran analytical assays for a biochemistry lab studying the dynamics and energetics of macromolecular interactions in signalling pathways.

ADDITIONAL EXPERIENCE

Bank of America Merrill Lynch, Healthcare Investment Banking — New York, NY
Investment Banking Summer Analyst — Jun – Aug 2019

- Supported financial analysis and industry research for healthcare clients; received a full-time return offer.

Mathematics & Science Tutor
Private tutoring, grades 6–12 — May 2017 – May 2021

- Taught pre-algebra through Calculus II, chemistry, and biology, plus SAT/ACT/DAT preparation.

TEACHING, MENTORING & SERVICE

- Member, Tempus AI Board of Education: organise cross-departmental educational workshops.
- Onboarding trainer for new Tempus scientists and external client teams.
- [Any PhD teaching / demonstrating / student supervision / peer review at Imperial?]
- Volunteer: Miriam's Kitchen (Washington, DC); Georgetown University Emergency Medical Response Service.

AWARDS & HONOURS

- Tempus AI "Stealth Mode" Award, 2022
- Georgetown University Dean's List (Spring 2017, Fall 2017); Second Honors (Fall 2016, Spring 2019)

CERTIFICATIONS

- Society for Immunotherapy of Cancer (SITC): Certificate in Cancer Immunotherapy
- HMX (Harvard Medical School): Genetics: Cancer Genomics & Precision Oncology; Genetics: Genetic Testing & Sequencing Technologies; Immunology: Immuno-oncology

TECHNICAL SKILLS

Methods	Causal inference, survival analysis, regression modelling, differential expression, unsupervised learning, deep learning, multi-omics integration (ICA, WGCNA, GSEA), real-world evidence, [imaging methods]
Languages	R (incl. package development), Python, SQL, Bash, MATLAB, LaTeX
Data & cloud	Google Cloud (BigQuery, GCS), AWS, dbt, Docker
Tools	Git/GitHub, Jira, Adobe Illustrator